14)Write a Code Using RAW Sockets to Implement Packet Sniffing

Program:

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/socket.h>
#include <arpa/inet.h>
#include <net/ethernet.h>

int main()
{
    int sockfd;
    unsigned char buffer[65536];

    sockfd = socket(AF_PACKET, SOCK_RAW, htons(ETH_P_ALL));

    if (sockfd < 0)
    {
        perror("Socket creation failed");
        return 1;
    }

    printf("Packet sniffing started...\n");

    while (1)
    {
        int size = recvfrom(sockfd, buffer,
                            sizeof(buffer), 0, NULL, NULL);

        if (size < 0)
        {
            perror("Packet receiving failed");
            break;
        }

        struct ethhdr *eth = (struct ethhdr *)buffer;

        printf("\nPacket received: %d bytes\n", size);

        printf("Source MAC: %02X:%02X:%02X:%02X:%02X:%02X\n",
               eth->h_source[0], eth->h_source[1],
               eth->h_source[2], eth->h_source[3],
               eth->h_source[4], eth->h_source[5]);

        printf("Destination MAC: %02X:%02X:%02X:%02X:%02X:%02X\n",
               eth->h_dest[0], eth->h_dest[1],
               eth->h_dest[2], eth->h_dest[3],
               eth->h_dest[4], eth->h_dest[5]);
    }

    close(sockfd);
    return 0;
}
```

Output :

```
mikey@mikey:~/exp12$ sudo ./sniff
Packet sniffing started...

Packet received: 230 bytes
Source MAC: 00:15:5D:1A:ED:B9
Destination MAC: 00:15:5D:7A:80:F3

Packet received: 230 bytes
Source MAC: 00:15:5D:7A:80:F3
Destination MAC: 00:15:5D:1A:ED:B9

Packet received: 230 bytes
Source MAC: 00:15:5D:1A:ED:B9
Destination MAC: 00:15:5D:7A:80:F3

Packet received: 230 bytes
Source MAC: 00:15:5D:7A:80:F3
Destination MAC: 00:15:5D:1A:ED:B9

Packet received: 230 bytes
Source MAC: 00:15:5D:1A:ED:B9
Destination MAC: 00:15:5D:7A:80:F3

Packet received: 230 bytes
Source MAC: 00:15:5D:7A:80:F3
Destination MAC: 00:15:5D:1A:ED:B9

Packet received: 42 bytes
Source MAC: 00:15:5D:7A:80:F3
Destination MAC: 00:15:5D:1A:ED:B9
```